

Introduction to Databases

Database, DBMS, RDBMS, NoSQL, and ACID

Lecture Notes

Course: Databases

Updated: 28/04/2026

Learning Objectives

After this lesson, learners will be able to:

1. Explain the concepts of **data** and **database**.
2. Describe the role of a **DBMS** in storing, retrieving, and managing data.
3. Describe the main components of a database.
4. Distinguish between **Relational Databases** and **NoSQL Databases**.
5. Explain the meaning of the **ACID** properties.
6. Identify real-world applications of databases in different domains.
7. Select suitable database types for selected technology domains.

Data and Information

Data

Data consists of raw facts, figures, or information that has not yet been organized and has little meaning when considered alone.

Examples of data:

- Customer name
- Phone number
- Student ID
- Exam score
- Transaction date
- Images, videos, and audio

Information

When data is processed, organized, and analyzed, it becomes meaningful information that supports decision-making.

What is a database?

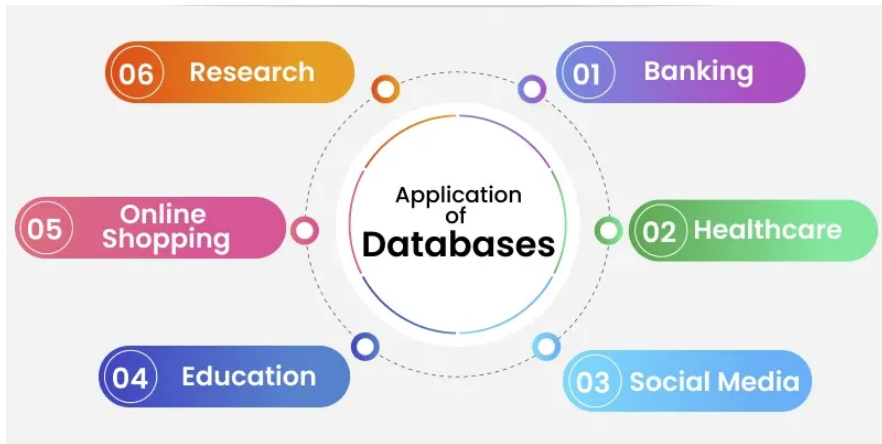
Definition

A database is a structured system used to **store, manage, retrieve, and update** data efficiently for multiple users and applications.

Databases are important because:

- **Scalability**: handles large volumes of data.
- **Data Integrity**: maintains accuracy through rules and constraints.
- **Security**: protects data through access rights and permissions.
- **Analytics**: supports analysis and decision-making.

Illustration: Why Databases Matter



Quiz: Data and Database Concepts

Question 1. What is data?

- A. Fully analyzed information
- B. Raw facts, figures, or unprocessed information
- C. A data management software package
- D. A type of programming language

Question 2. What is a database mainly used for?

- A. Designing user interfaces
- B. Storing, managing, and retrieving data
- C. Only for writing Python programs
- D. Only for storing images

Quiz: Data and Database Concepts (cont.)

Question 3. Which feature helps a database handle large volumes of data?

- A. Security
- B. Scalability
- C. Isolation
- D. Durability

How Databases Work

Databases organize and store information in structured or unstructured formats, allowing users to easily:

- Access, search for, and retrieve data.
- Update, delete, and manage data.
- Manage access permissions.

The Central Role of a DBMS

A DBMS is a software layer between users and raw data. Users interact with the DBMS through queries or applications instead of directly handling physical storage details.

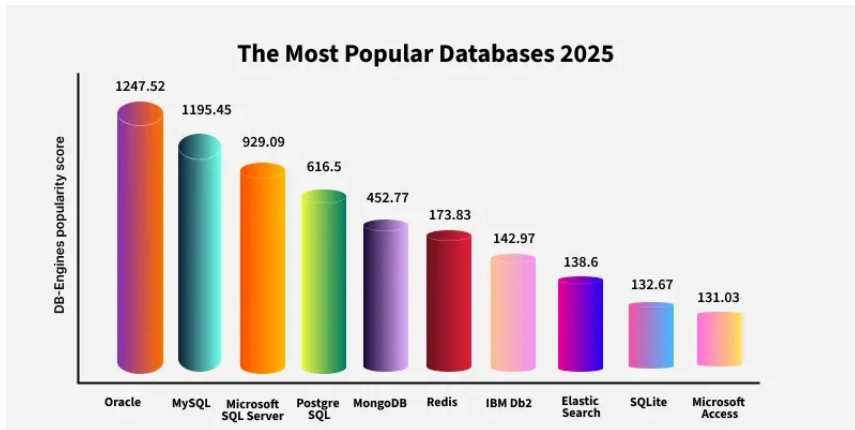
Request Processing in a DBMS

1. A user or application sends a request to the DBMS.
2. The DBMS checks syntax, verifies access rights, and optimizes the query.
3. The DBMS accesses relevant data from tables, indexes, or storage files.
4. The DBMS returns results to the user or application.

The DBMS also provides:

- Data backup and recovery.
- Performance optimization.
- Transaction management.
- Access control.
- Concurrent multi-user management.

Illustration: Request Processing in a DBMS



Quiz: How Databases Work

Question 1. What role does a DBMS play in a database system?

- A. It is an intermediary layer between users and data
- B. It is a hardware device used to store data
- C. It is a programming language that replaces SQL
- D. It is software used only for drawing charts

Question 2. When a user submits a query, what does the DBMS usually do?

- A. Ignores the query
- B. Processes the query, finds relevant data, and returns results
- C. Always deletes data before returning results
- D. Only saves the query into a text file

Quiz: How Databases Work (cont.)

Question 3. Which of the following is an important DBMS function?

- A. Data backup and recovery
- B. Only playing music
- C. Only image editing
- D. Only compiling C++ programs

Components of a Database

A database consists of multiple components that work together to store, organize, manage, and retrieve data efficiently.

Component	Main role
Data	The actual information stored in the database.
Schema	The blueprint that describes how data is organized.
DBMS	Software that manages database operations.
Query	Instructions used to retrieve, add, update, or delete data.
Users	Individuals or systems that interact with the database.

Illustration: Database Components

Relational databases



Table-based

Non-relational databases



Key-Value



Wide-column



Graph



Document

Component 1: Data

Concept

Data is the actual information stored in the database.

Data may include:

- Text, numbers, and dates.
- Images, files, audio, and video.
- Sensor data or system-generated data.

Example

In a student management system, data may include student ID, full name, date of birth, class, exam scores, and academic status.

Component 2: Database Schema

Schema

A **schema**, or **database schema**, is the blueprint that describes how data is organized.

A schema may describe:

- Tables and fields.
- The data type of each field.
- Primary keys and foreign keys.
- Relationships between tables.
- Data constraints.

Example of a Database Schema

In a sales management database, the schema may include the following tables:

- Customers: stores customer information.
- Orders: stores order information.
- Products: stores product information.
- OrderDetails: stores line-item details for each order.

Meaning

A schema tells the system how data is organized, how tables are connected, and which constraints the data must satisfy.

Component 3: Database Management System

DBMS

A **DBMS** is software that manages database operations.

A DBMS supports functions such as:

- Storing, retrieving, updating, and deleting data.
- Managing security and users.
- Transaction management.
- Data backup and recovery.

Examples of DBMSs

Some popular database management systems:

- MySQL.
- PostgreSQL.
- Oracle.
- Microsoft SQL Server.
- MongoDB.

Remember

A DBMS does not only store data; it also controls how data is accessed, protected, updated, and recovered after failures.

Illustration: Database Management System



Component 4: Query

Query

A **query** is an instruction used to retrieve, add, update, or delete data in a database.

In relational databases, queries are usually written in **SQL**.

```
SELECT * FROM Students  
WHERE major = 'Data Science';
```

The query above retrieves the list of students majoring in *Data Science*.

Component 5: Users

Users are individuals or systems that interact with a database.

Common user groups include:

- Database administrators.
- Developers.
- Data analysts.
- End users.
- Software applications.
- Automated systems.

Authorization

Each user group may have different access privileges depending on its role.

Two Common Database Groups

Relational Database

- Organizes data into tables.
- Has rows, columns, primary keys, and foreign keys.
- Commonly uses SQL.
- Suitable for structured data.

NoSQL Database

- More flexible in data models.
- Suitable for unstructured or semi-structured data.
- Supports horizontal scalability well.
- Includes document, key-value, wide-column, and graph databases.

Relational Databases

A **Relational Database** organizes data into tables. Each table includes:

- **Rows/Records:** specific records.
- **Columns/Fields:** attributes of the records.

StudentID	FullName	Major	GPA
S001	Nguyen An	Data Science	3.4
S002	Tran Binh	AI	3.7
S003	Le Chi	Information Systems	3.2

Characteristics of Relational Databases

- **Strict schema:** tables, columns, data types, and constraints are predefined.
- **Primary key:** uniquely identifies each record in a table.
- **Foreign key:** represents relationships between tables.
- **ACID:** ensures reliable transactions.
- **Structured data:** suitable for accounting, banking, student management, and sales.

Examples of RDBMSs: MySQL, PostgreSQL, Oracle, and Microsoft SQL Server.

Quiz: Relational Databases

Question 1. How do relational databases mainly organize data?

- A. Graphs
- B. Tables with rows and columns
- C. Audio files
- D. Blockchain

Question 2. What is a primary key used for?

- A. Uniquely identifying each record in a table
- B. Deleting the entire database
- C. Encrypting images
- D. Creating a user interface

Quiz: Relational Databases (cont.)

Question 3. Which of the following is an RDBMS?

- A. PostgreSQL
- B. Redis
- C. Neo4j
- D. IPFS

NoSQL Databases

NoSQL stands for *Not Only SQL*. NoSQL databases are designed to handle unstructured or semi-structured data such as:

- Text, images, and videos.
- Sensor data.
- Log system.
- Social media data.
- Real-time data.

Unlike RDBMSs, NoSQL databases do not necessarily store data in fixed tables.

Main Types of NoSQL Databases

Type	Description	Example
Document	Stores data as JSON-like documents	MongoDB
Key-value	Stores key-value pairs for fast lookup	Redis
Wide-column	Stores data by columns; suitable for big data	Cassandra
Graph	Stores nodes and relationships between nodes	Neo4j

Quiz: NoSQL Databases

Question 1. What does NoSQL stand for?

- A. No Software Query Language
- B. Not Only SQL
- C. New Online SQL
- D. Normal SQL

Question 2. Which NoSQL type stores data as key-value pairs?

- A. Document database
- B. Key-value store
- C. Graph database
- D. Relational database

Quiz: NoSQL Databases (cont.)

Question 3. Neo4j is an example of which type of database?

- A. Graph database
- B. Key-value store
- C. RDBMS
- D. File system

What Is ACID?

ACID consists of four important properties that make database transactions reliable, accurate, and safe.

Property	Meaning
Atomicity	A transaction must be executed completely or not at all.
Consistency	The database moves from one valid state to another valid state.
Isolation	Concurrent transactions do not incorrectly affect one another.
Durability	After a transaction is completed, changes are stored permanently.

ACID Example: Bank Transfer

When transferring money from account A to account B:

1. Debit account A.
2. Credit account B.

- **Atomicity**: if step 2 fails, step 1 must be rolled back.
- **Consistency**: the balance remains valid after the transaction.
- **Isolation**: concurrent transactions do not corrupt the data.
- **Durability**: a successful transaction must be stored permanently.

Quiz: ACID Properties

Question 1. Which properties does ACID include?

- A. Accuracy, Control, Integrity, Design
- B. Atomicity, Consistency, Isolation, Durability
- C. Access, Cache, Index, Data
- D. Application, Cloud, Internet, Database

Question 2. Which property ensures that a transaction is executed completely or not at all?

- A. Atomicity
- B. Consistency
- C. Isolation
- D. Durability

Quiz: ACID Properties (cont.)

Question 3. Which property ensures that data remains stored after a transaction has completed?

- A. Atomicity
- B. Consistency
- C. Isolation
- D. Durability

Real-World Applications of Databases

Domain	Commonly managed data
Banking	Transactions, accounts, loans, and customer information.
Transportation	Bookings, schedules, routes, vehicles, and passengers.
Education	Student records, grades, schedules, and tuition fees.
Retail	Inventory, orders, customers, promotions, and revenue.
Social media	Users, posts, messages, images, videos, and interactions.

Quiz: Database Applications

Question 1. In banking, what are databases commonly used to store?

- A. Transactions and accounts
- B. Only desktop wallpapers
- C. Only program source code
- D. Only audio files

Question 2. In education, what can databases be used to manage?

- A. Student records and grades
- B. Computer fan speed
- C. TV control signals
- D. Interface colors

Quiz: Database Applications (cont.)

Question 3. In retail, what can databases help manage?

- A. Inventory and orders
- B. Only personal videos
- C. Only router IP addresses
- D. Only music players

Databases in Technology Domains

- **Web Development:** MySQL, PostgreSQL, MongoDB, Firebase.
- **Mobile Development:** SQLite, Realm, Firebase Realtime Database.
- **DevOps:** PostgreSQL, Redis, InfluxDB, Cassandra.
- **Data Engineering:** HDFS, Cassandra, Redshift, BigQuery.
- **Data Science:** PostgreSQL, MongoDB, Hive, Snowflake.
- **AI:** MongoDB, Cassandra, BigQuery, AWS S3.
- **Cloud:** Aurora, Cloud Spanner, Azure Cosmos DB.
- **Blockchain/Web3:** BigchainDB, IPFS, Ethereum, Chainlink.

Quiz: Databases by Technology Domain

Question 1. Which database is often used in mobile applications because it is lightweight and suitable for local storage?

- A. SQLite
- B. Oracle RAC
- C. HDFS
- D. Ethereum

Question 2. Which domain is Google BigQuery commonly suitable for?

- A. Data engineering and big data analytics
- B. Only image editing
- C. Small offline game programming
- D. Keyboard control

Quiz: Databases by Technology Domain (cont.)

Question 3. Which field is IPFS commonly associated with?

- A. Blockchain/Web3
- B. Traditional classroom management
- C. Only personal desktop applications
- D. The Windows operating system

Comparison: Relational Database vs. NoSQL

Criterion	Relational Database	NoSQL Database
Structure	Tables, rows, columns	Documents, key-value, wide-column, graphs
Schema	Usually fixed and strict	More flexible
Language	SQL	Depends on the DBMS
Relationships	Primary keys, foreign keys, joins	Depends on the data model
Scalability	Usually strong in vertical scaling	Usually strong in horizontal scaling
Suitable for	Structured data	Unstructured/semi-structured data
Examples	MySQL, PostgreSQL, Oracle	MongoDB, Redis, Cassandra, Neo4j

Review Questions: Multiple Choice

Question 1. What is a database?

- A. A system used to store, manage, and retrieve data
- B. A web browser
- C. An operating system
- D. An input device

Question 2. What is a DBMS?

- A. Software for managing databases
- B. A type of monitor
- C. A social network protocol
- D. Software used only for drawing images

Review Questions: Multiple Choice (cont.)

Question 3. Which component describes table, field, and relationship structures?

- A. Schema
- B. User
- C. Query
- D. Image file

Question 4. What do relational databases organize data into?

- A. Tables
- B. Video
- C. Voice messages
- D. An unstructured random bit string

Review Questions: Multiple Choice (cont.)

Question 5. What type of data is NoSQL suitable for?

- A. Unstructured or semi-structured data
- B. Only rigid tabular data
- C. Only small integers
- D. Only data printed on paper

Question 6. What does the Isolation property in ACID mean?

- A. Concurrent transactions do not interfere with each other incorrectly
- B. Data is never stored
- C. Every table must have an illustration
- D. Queries must be written in Vietnamese

Review Questions: Multiple Choice (cont.)

Question 7. MongoDB is an example of which database type?

- A. Document database
- B. Graph database
- C. Key-value store
- D. Traditional relational database

Question 8. What is Redis commonly used for?

- A. Caching or fast key-value lookup
- B. Designing presentation slides
- C. Drawing 3D models
- D. Only storing long videos

Review Questions: Multiple Choice (cont.)

Question 9. What kind of problems is Neo4j suitable for?

- A. Relationship and graph analysis
- B. Calculating payroll on paper
- C. Storing simple local music files
- D. Only editing text

Question 10. In AI, what can databases be used for?

- A. Storing training datasets and model feedback
- B. Only setting application background colors
- C. Completely replacing machine learning algorithms
- D. Only managing the mouse and keyboard

Review Questions: Short Answer

1. Distinguish between data and information.
2. Explain the role of a DBMS in a database system.
3. Why are relational databases suitable for structured data?
4. Why is NoSQL suitable for big data, unstructured data, or rapidly changing data?
5. Explain the meaning of the four ACID properties using a real-world example.

Practice Exercises

Exercise 1

A university wants to build a student management system including student information, classes, courses, grades, and tuition fees. Propose a suitable database type and explain your choice.

Exercise 2

A social network needs to store user profiles, posts, comments, images, videos, and friendship relationships. Propose one or more suitable database types.

Further Practice Exercises

Exercise 3

An e-commerce application needs to store products, orders, customers, payments, and purchase history. Should it combine a relational database and NoSQL? Why?

Exercise 4

An AI system needs to store training datasets consisting of text, images, and real-time user feedback. Propose a suitable data storage architecture.

Lesson Summary

- Data is raw information; databases help organize and manage data efficiently.
- A DBMS is intermediary software that helps users and applications interact with data.
- A database consists of data, schema, DBMS, queries, and users.
- RDBMSs organize data in tables and are suitable for structured data.
- NoSQL is suitable for unstructured data, semi-structured data, and systems requiring horizontal scaling.
- ACID helps ensure that database transactions are reliable, accurate, and safe.

Key Terms

- Data, Information, Database
- DBMS, Schema, Query, SQL
- Relational Database, RDBMS
- NoSQL, Document Database
- Key-Value Store
- Wide-Column Database
- Graph Database
- ACID
- Atomicity, Consistency
- Isolation, Durability
- Data Engineering, Data Science
- AI, Cloud, Blockchain, Web3

Thank you!

Discussion and Questions